Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/03/2004
PI Name:	Mazyck, David W. Ph.D.		
Project Title:	Magnetically Agitated Photocatalytic Reactor for Water Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmazyck@ufl.edu	**Fax:**	FY 352-392-3076
PI Organization Type:	UNIVERSITY	**Phone:**	352-846-1039
Organization Name:	University of Florida		
PI Address 1:	306 black hall		
PI Address 2:	PO Box 116450		
PI Web Page:	http://www.ees.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-6450	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	9-CAN-00-BH
Start Date:	06/01/2002	**End Date:**	06/14/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chadik, Paul (University of Florida) Wu, CY (University of Florida) Powers, Kevin (University of Florida) Sigmund, Wolfgang (University of Florida) Meisel, Mark (University of Florida)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA has identified a number of water treatment options that have shown promise in space. The water recovery system on any particular mission will be a collection of individual treatment units, with each treatment unit focusing on a select group of contaminants. The goal of this research is to develop a post processor for the destruction of organic compounds and microbes that may not be removed in water subsystems prior to the MAPR. The overall objective is to develop a microgravity-compatible, compact reactor that is safe and reliable that provides product water that meets or exceeds NASA's potable water quality requirements. Furthermore, this innovative technology has a reduced ESM because of its compactness and ability to function reliably without the extensive use of consumables (i.e., infrequent replacement of the UV lamps). The technology proposed herein is applicable for implementation on board the ISS, in-flight, or for planetary (e.g., Lunar, Mars, etc.) missions. The assessment phase of the project will quantify the reactor removal efficiency with respect to pH, conductivity, total organic carbon (TOC), and trace organics. Optimize the MAPR with respect to particle design and loading, number of UV lamps, and UV wavelength. Post processing of water from prior subsystems is crucial to ensure crew health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Magnetically Agitated Photocatalytic Reactor (MAPR) under development at the University of Florida is well suited for many water purification applications including recovering water during missions in space, removing contaminants from water in the home, providing drinking water onboard ocean vessels, supplying troops with drinking water during military operations, and providing safe water for disaster response when water is desperately required but not available. As sunlight can be used as the light source, this technique offers a unique and affordable solution to water quality problems that face developing countries. In addition to water applications, the MAPR could be designed for air purification applications including use in the airline industry, underground mining, or automobiles. Traditionally water has been purified of harmful contaminants by activated carbon adsorption or photocatalytic oxidation. Although both are effective, activated carbon has a finite adsorption capacity and requires replacement or regeneration. Photocatalytic slurries may encounter problems such as separation of photocatalysts from recovered water. The advantage of using magnetic substrates is that they can be agitated by an external magnetic field to provide enhanced mixing of the photocatalyst particles with pollutants, and to enhance exposure to UV light. In addition, because the magnetic substrates are controlled by a magnetic field, the recovered water will be free of the composite particles. Faster destruction rates allow for compact treatment devices, which are critical to the applications listed above. Furthermore, this innovative technology is attractive because of its compactness and ability to function reliably without the extensive use of consumables (i.e., very infrequent replacement of UV lamps and media).
Task Progress:	Photocatalytically active magnetic particles were synthesized using a variety of techniques. Coating durability and stability were the primary obstacles to overcome. After many attempts, a working, durable, photocatalytically active composite particle was synthesized using a sol-gel technique. These TiO2-silica-baruim ferrite composites were composed of a hydrolyzed alkoxide (TBOT) to create the TiO2 layer, a silica intermediate layer, and ball-milled barium ferrite as the core. The particles were heat treated to form the anatase crystalline phase and had a mean particle size of 2.2 µm, 11 wt% titania loading, and a BET surface area of approximately 27 m2/g . A simple model was developed and used to estimate the system’s resonant frequency as a function of particle magnetization and magnetic field gradient. This model accurately predicted the range of the resonant frequency. A frequency range within the predicted bounds was tested to determine the optimum agitation frequency using light scattering and dye degradation experiments. Both experiments correlated in returning an optimum agitation frequency of 50 Hz. The composite particles were then tested in a flow through system to quantify phenol degradation. Particle mass, UV wavelength, and the number of UV lights were varied to determine the optimum system parameters. Optima for phenol degradation included a 2.5 g/L particle loading and three 8-watt 312 nm UV lamps. At the conclusion of the experiments, the magnetic particles were easily removed from the solution by applying an external magnetic field.
Bibliography Type:	Description: (Last Updated: 10/04/2004)
Articles in Peer-reviewed Journals	Lee, S.W.; Driewga, J.; Wu, C.Y.; Mazyck, D.; Sigmund, W.M. "Anatase TiO2 nanoparticle coating on barium ferrite using titanium bis-ammonium lactato dihydroxide and its use as a magnetic photocatalyst." Chemistry of Materials 16(6):1160-1164, Mar 2004. , Mar-2004